温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Statistical Inference III
Hypothesis Testing

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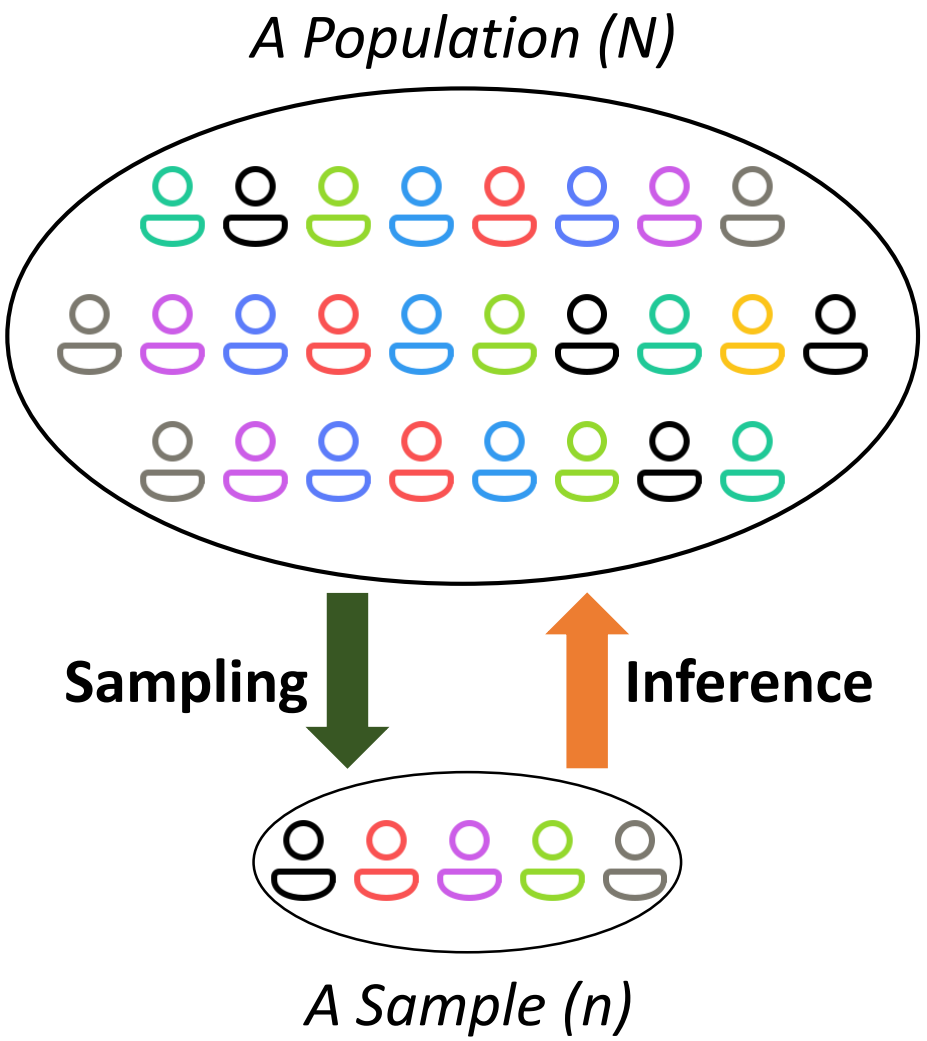


Statistical inference

- Focuses on drawing conclusions about populations from samples.

- Includes estimation of population parameters and hypothesis testing, which involves drawing conclusions about the value of the parameters of one or more populations.

Review: Sampling Distribution of the Sample Mean ($\bar{x}$)

- What We Learned
  - The sample mean $\bar{x}$ is a **random variable**.
  - The sampling distribution of $\bar{x}$ describe how $\bar{x}$ behaves across all possible samples of size n.
  - **<u>CLT:</u>** If the population is normal (any n) **<u>or</u>** the sample size n is large enough (any population distribution), the sample mean $\bar{x}$ follows approximately a normal distribution.

$$\sqrt{n}(\bar{x}-\mu) \xrightarrow{d} N(0,\sigma^2) \Leftrightarrow \bar{x} \xrightarrow{d} N(\mu, (\frac{\sigma}{\sqrt{n}})^2) \Leftrightarrow \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

convergence in distribution or convergence in law

- Impact of Sample Size n
  - Intuition: As n increases → samples become *<u>more similar to each other</u>* and *<u>to the population</u>*.
  - Sampling distribution: As n increases → the sampling distribution **narrows**, standard error $\frac{\sigma}{\sqrt{n}}$ **decreases**, yielding more **precise** estimate of population mean μ.

For finite population and $\frac{n}{N} > 0.05$:

$$\sigma_{\bar{x}} = \sqrt{\frac{N-n}{N-1}}\frac{\sigma}{\sqrt{n}}$$

VS.

For infinite population or finite population but $\frac{n}{N} > 0.05$:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Review: Understand Confident Interval

- Where does the function come from?
- Why can we say (for example) a 90% confidence level means we are 90% confident that the interval includes the population mean μ?

Critical Value **Std**

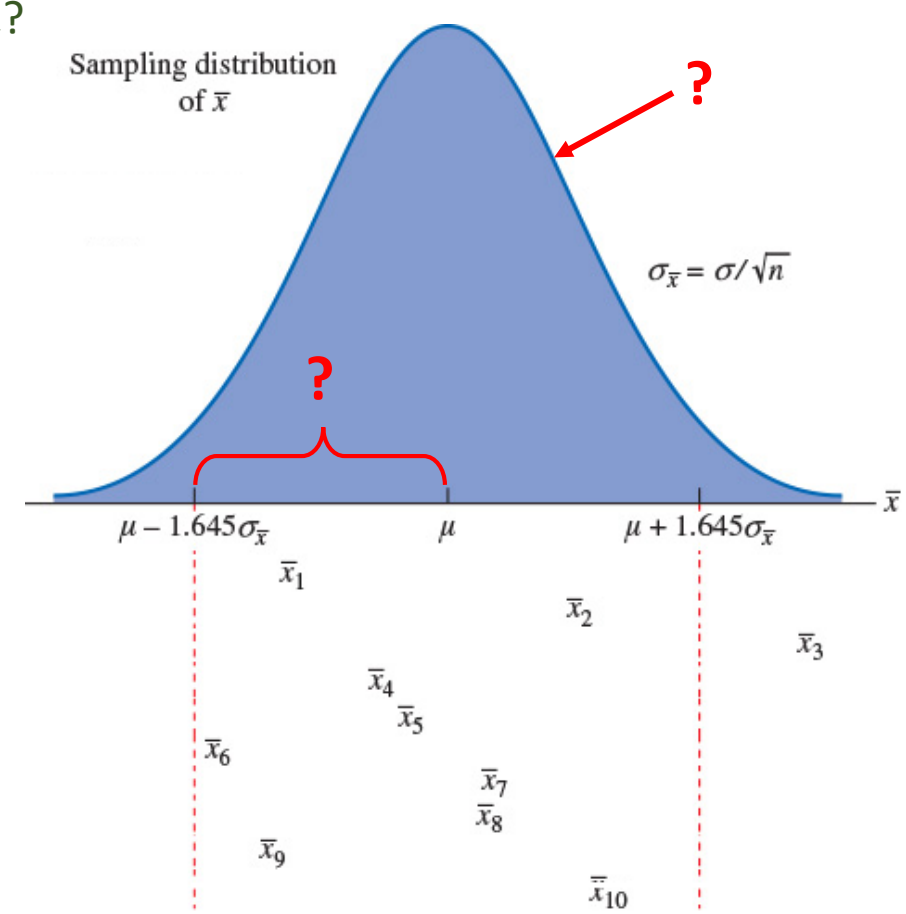
$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Significance Level

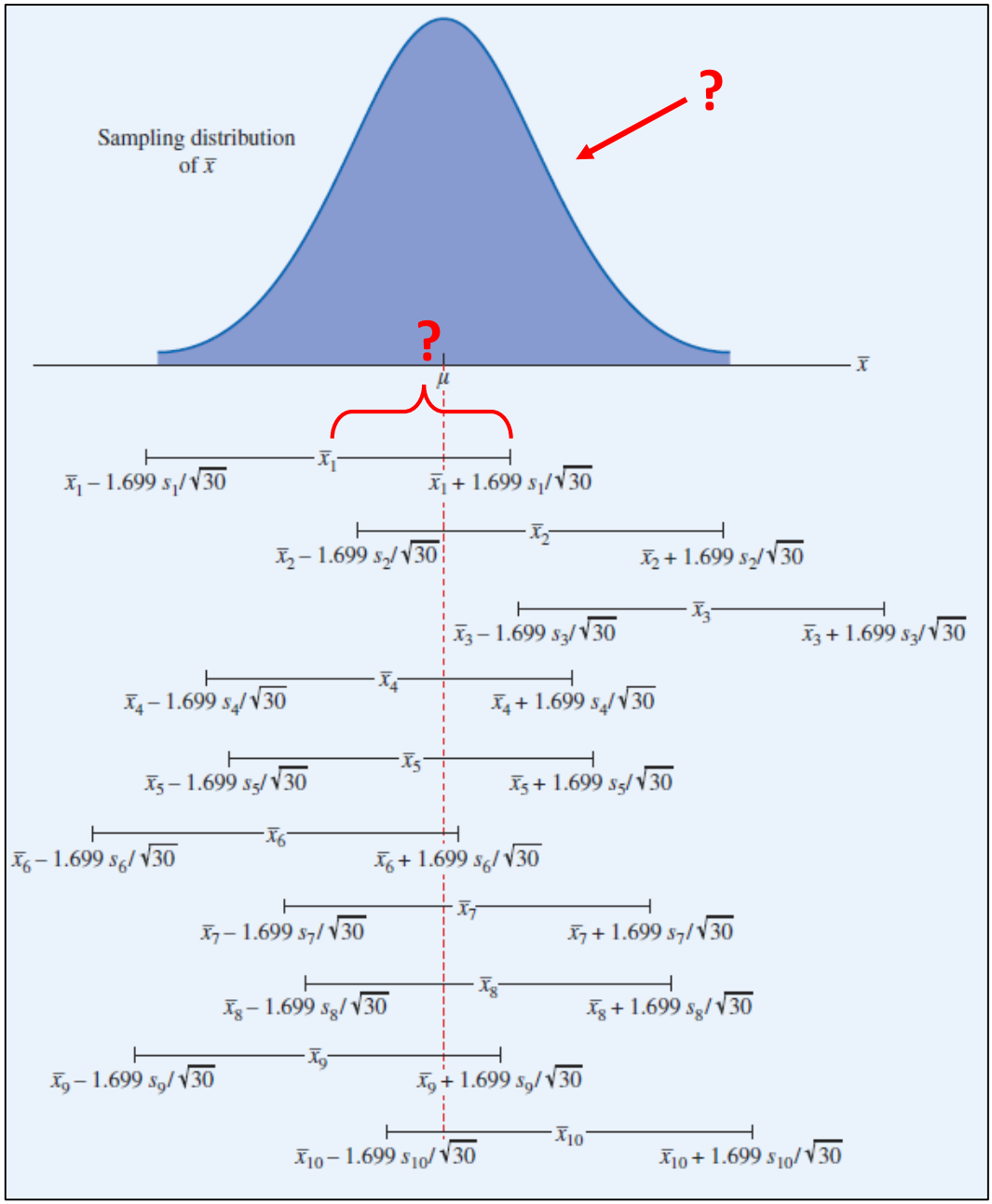
$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$\bar{x} \pm t_{n-1} \frac{s}{\sqrt{n}}$$

When to use?



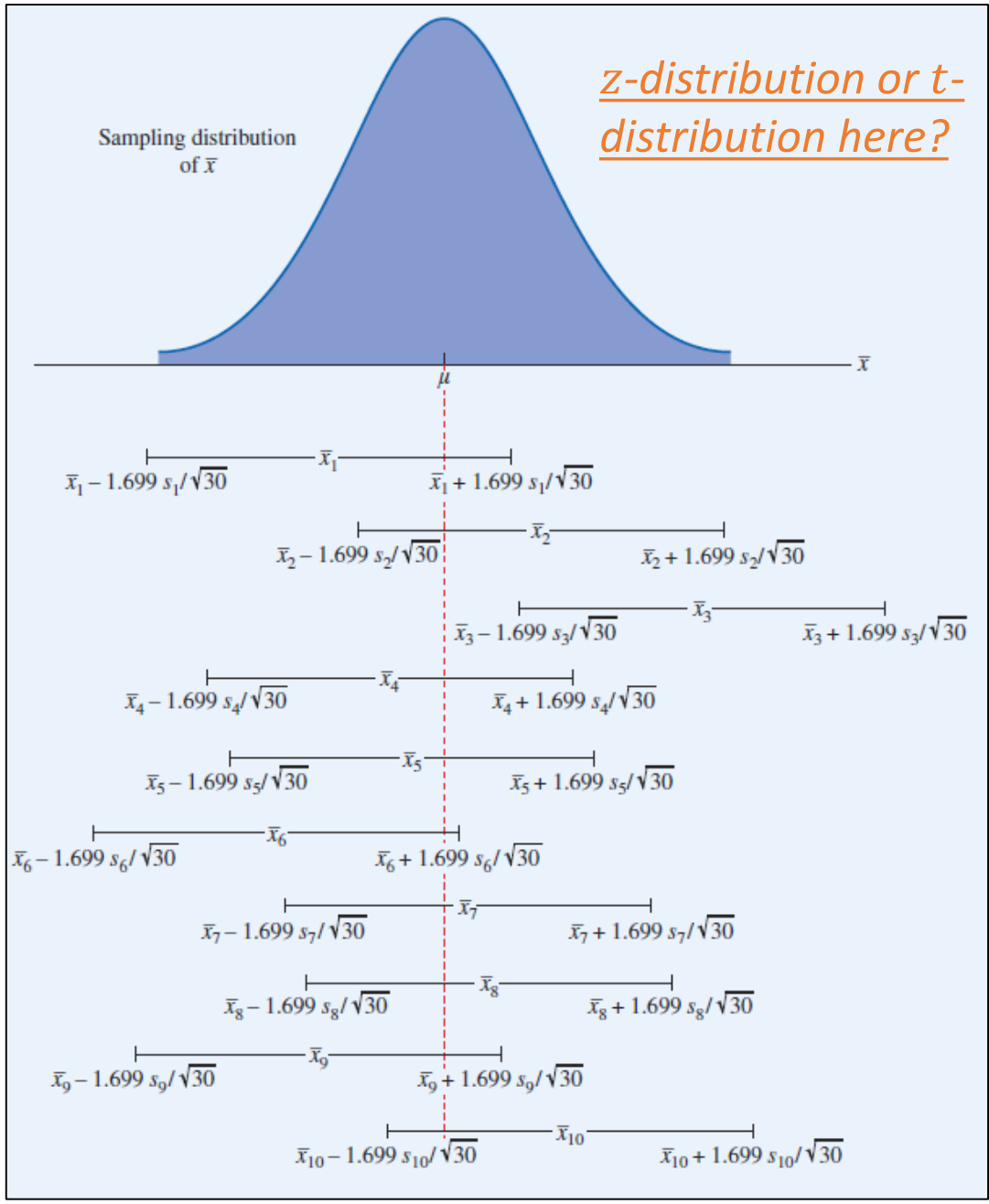
$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \qquad \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$



z-distribution vs. t-distribution

Review: Interpretation of Confident Interval

- Approximately 90% of all the intervals constructed using $\bar{x} \pm 1.699(s/\sqrt{30})$ will contain the population mean.
- We are approximately 90% confident that the interval $[\$70{,}778, \$72{,}850]$ includes the population mean μ.
- This interval has been established at the 90% confidence level.
- <u>95% confidence interval does not have a 95% probability of containing the population mean μ.</u>
- Once constructed, a confidence interval will either contain the population parameter or not contain the population parameter.
- If we *take several independent samples* of the same size from our population and construct a 95% confidence interval for each of these samples, we would expect *95% of these confidence intervals to contain the mean μ*.



Camm Ch7, p342, Figure 7.11

Example: Confident Interval for a Population Proportion

- **Scenario:** A survey of 200 WKU students finds that 60 prefer mobile ordering at the campus café.
  1) Estimate the population proportion (p) of all students who prefer mobile ordering, with **95%** confidence.

- Step 1. Compute the sample proportion $\hat{p} = \frac{60}{200} = 0.30$
- Step 2. Find the standard error (SE) $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.3(1-0.3)}{200}} = 0.0324$
- Step 3. Identify the z-value for 95% confidence $z_{0.025} = 1.96$
- Step 4. Compute the confidence interval $\hat{p} \pm z \times SE = 0.30 \pm 1.96(0.0324)$
  $\Rightarrow$ (0.236, 0.364)

  2) What is the interpretation of this 95% confident interval?

We are **95% confident** that between **23.6% and 36.4%** of all WKU students prefer mobile ordering.

Opening Hook – Chatbot vs. Human Help Desk

- Context
  - The WKU Library recently launched an **AI chatbot** to answer student questions. Before, all inquiries were handled by human staff, and student satisfaction scores averaged 8.0 out of 10.
  - After the chatbot launch, the library surveyed **40** students who used it:
    - Average rating: **7.6**
    - Some said it's fast and convenient.
    - Others missed the "human touch."
- Now, the library faces a question:
  - Is the chatbot performing just **as well as** the humans — or **is the difference real**? Or could this lower average (7.6 vs. 8.0) just be **random** because we only asked 40 students (sampling error)?

Opening Hook – Chatbot vs. Human Help Desk

- Discussion
  1) What is the population?
  2) If we surveyed **another** 40 students, do you think we'd get the **same** average?
  3) How much difference would you call "meaningful"? 0.1? 0.5? 1.0 point?
  4) What if we asked 400 students instead of 40 — would we feel **more confident** about your conclusion?
  5) What if the chatbot's average were **7.9** instead of **7.6** — would your judgment change?
  6) Now imagine two **different** groups of 40 students both gave an average of 7.6, but:
     - Group A's ratings ranged from 7.5 to 7.7, with std 0.5 (very consistent).
     - Group B's ratings ranged from 4 to 10, with std 2 (highly variable).

     → In which case would you feel **more confident that the chatbot truly scores below 8.0**?
  7) How certain do we need to be before deciding the chatbot *really* performs worse – 90% sure? 95% sure?

Opening Hook – Chatbot vs. Human Help Desk

Spirit of Hypothesis Testing

- These questions capture the essence of hypothesis testing — using sample data to decide whether an observed difference is **real** or could simply be due to **random chance**.

- The sample size, the variability (standard deviation), and how confident we want to be all affect *how strong our evidence must be before we conclude a real difference exists*.

Hypothesis Testing

- Purpose
  - To determines whether a statement about a population parameter **should be rejected or not** based on sample data.
  - It involves drawing inferences about two contrasting propositions (each called a <u>hypothesis</u>) relating to the value of one or more population parameters.
- Procedure
  1) Develop the null and alternative hypotheses (H_0 vs. H_1 or H_a)
  2) Specify the level of significance
  3) Collect the sample data and calculate a test statistic
  4) Determine the decision rule on which to base a conclusion
     - Critical value, rejection region, p-value
  5) Apply the decision rule and draw a conclusion in the context of the application

Step 1: Develop Hypotheses H_0, H_1

- The hypothesis testing procedure uses data from a sample to test the validity of the two competing statements indicated by H_0 and H_1 or H_a.
- $\boldsymbol{H_0}$: null hypothesis, is a statement (a tentative conjecture) about a population parameter, describes an existing theory, "default", status quo assumption (read as "H zero" or "H naught")
- $\boldsymbol{H_1}$: alternative hypothesis, is the opposite of what is stated in the null hypothesis H_0, the complement of H_0
- Using sample data, we either:
  - Reject H_0 and **conclude** the sample data provides sufficient evidence to support H_1, or
  - Fail to reject H_0 and **conclude** the sample data does not support H_1.

Why not "accept H_0?"

No proof that H1 is true or false

Step 1: Develop Hypotheses H_0, H_1

- When setting up hypotheses, ask:
  - What is the **purpose** of collecting the sample?
  - What **conclusions** do we hope to draw from the data?
- Sometimes it's easier to start with the alternative hypothesis (what we want to show); other times, begin with the null hypothesis (the current belief or status quo).
- Examples
  - H_1 as a research hypothesis
  - H_0 as an assumption to be challenged

Step 1: Develop Hypotheses H_0, H_1

H_1 as a research hypothesis

- Example
  - A computer-aided design (CAD) software company serving the aerospace industry receives many technical support calls. The average response time has historically been at least 25 minutes. The company upgraded its support system to reduce response time. A sample of **44** customer calls was collected to evaluate whether the new system is effective.

$$H_0\colon\ \mu \geq 25 \qquad H_1\colon\ \mu < 25$$

- Key Takeaways
  - In this example, the company would like to conclude that new system is effective or the response time is less than 25 minutes – this becomes the alternative hypothesis ($H_1\colon\ \mu < 25$). The null hypothesis represents the opposite or status quo: $H_0\colon\ \mu \geq 25$.
  - When testing a research claim, it often helps to start from the alternative hypothesis — the idea you want to support.
  - We conclude that the research claim is supported *only if the sample data provide enough evidence to reject* H_0.

Step 1: Develop Hypotheses H_0, H_1

H_0 as an Assumption to be Challenged

- Example
  - A bottle label states it contains **200g net weight**. You collect test data to determine whether this claim is supported by evidence or **if the true/population net weight is 200g**.

$$H_0\colon \mu = 200 \qquad H_1\colon \mu \neq 200$$

- Key Takeaways
  - In this example, we start by assuming the label is correct — that the population mean net weight is $H_0\colon \mu = 200\text{g}$. The alternative hypothesis challenges this assumption.
  - In many cases, we begin with a belief or claim about a population parameter and then test whether the data provide enough evidence to reject that assumption.
  - When testing a stated claim (like a product label), it is usually best to formulate the null hypothesis first.

Step 1: Develop Hypotheses H_0, H_1

- The equality part of the hypotheses is always in the null hypothesis. Why?
- In general, a hypothesis test about a population mean (μ) takes one of three forms, where μ_0 is the hypothesized value (same logic for other population parameters):

One-tailed (Lower-tail)	**One-tailed (Upper-tail)**	**Two-tailed**
H_0: $\mu \geq \mu_0$	H_0: $\mu \leq \mu_0$	H_0: $\mu = \mu_0$
H_1: $\mu < \mu_0$	H_1: $\mu > \mu_0$	H_1: $\mu \neq \mu_0$

- The choice depends on the research question or claim you want to test:
  - Use two-tailed (H_1: $\mu \neq \mu_0$) when testing for any difference (≠).
  - Use upper-tailed (H_1: $\mu > \mu_0$) when testing if the true mean is greater than a specific value.
  - Use lower-tailed (H_1: $\mu < \mu_0$) when testing if the true mean is **less** than a specific value.
  - In short: **the direction of your claim determines the direction of the test.**

Step 2: Significance Level α

Suppose the WKU Library is willing to accept a **1% risk** of making an incorrect conclusion when testing the AI chatbot's performance claim.

- Because we rely on sample data, there is always a chance of making a wrong conclusion due to **sampling error**.
- Every hypothesis test has four possible outcomes:

Any question on this table?

		Population Condition	
		H_0 True	**H_0 False**
Conclusion you made	**Fail to reject H_0**	Correct Inference (true negative) (probability = $1 - \alpha$)	Type II Error (false negative) (probability = β)
	Reject H_0	Type I Error (false positive) (probability = α)	Correct Conclusion (true positive) (probability = $1 - \beta$)

Step 2: Significance Level α

Why "Reject H_0" ("Fail to Reject H_0") Feels Positive (Negative)?

- "Reject H_0" – **Positive / Action-Oriented**
  - Suggests a new finding or improvement — evidence shows something has changed.
  - Recall: H_0 represents the default or status quo; researchers usually *hope to reject it*.
  - Example: a new drug works, a new system performs better, a new method is effective.

- "Fail to reject H_0" – **Negative / Inconclusive**
  - Indicates there is not enough evidence to support the new claim.
  - We remain with the status quo, but we *do not prove it true — only that we can't reject it*.

Step 2: Significance Level α

A Legal Analogy for Hypothesis Testing

- Null hypothesis (H_0)
  - The default assumption – current belief, status quo, "innocent until proven guilty."
  - It is the claim we want to challenge.
- Alternative Hypothesis (H_1)
  - What we want to show or conclude – the "guilty" claim.
- Analogy
  - In court, a defendant is innocent until proven guilty.
    - H_0: The defendant is innocent.
    - H_1: The defendant is guilty.
- If the evidence (sample data) strongly indicates the defendant is guilty, we reject H_0 and conclude guilt.
- If the evidence is not sufficient to indicate guilty, we fail to reject the "innocent" hypothesis (H_0). However, note that we have not proven guilt or innocence!

Step 2: Significance Level α

A Legal Analogy for Hypothesis Testing (Continued)

- Why do we say "innocent until proven guilty" (无罪推定) rather than "guilty until proven innocent" (有罪推定)?

  → 疑罪从无 Safer to assume innocence, require strong evidence before punishment

  → Better to release a suspect than wrongly convict an innocent person (宁可错放，不可错判)
- Which is easier: to prove guilty or to prove innocence?
- Which is easier: to show you did something or prove you didn't?
- In hypothesis testing, the null hypothesis (H_0) plays the same role as *"innocent until proven guilty."* → We assume H_0 is true until there's strong evidence to reject it.

Step 2: Significance Level α

- Example: "I will not detect, criticize, or penalize AI use because of ***false positives***."
  - What is the null hypothesis (H_0) here?
    - The student did not use AI.
  - What is "false positive" here?
    - Accusing a student of using AI when they actually did not.
  - What is "false negative" here?
    - Believing a student did not use AI when they actually did.
  - Which error is more serious: 误判 (false positive) or 漏判 (false negative)?
    - False positive is more serious. Wrongly accusing a student of using AI when they did not can **damage trust, reputation, and fairness**.
    - A false negative – missing an actual AI user – is **less ideal**, but it doe **not harm innocent students** directly.
  - Why should we be more cautious about wrongly rejecting the null hypothesis (type I error)?
    - Because rejecting H_0 means we're claiming a difference or wrongdoing exists.
    - If we reject H_0 incorrectly, we make a Type I error, accusing or concluding something untrue.
    - In many real situations (law, medicine, education), a false accusation or wrong conclusion can cause **greater harm** than missing a real effect.
    - *So we set a small significance level (α) to keep the risk of wrongly rejecting H_0 very low.*

Generally, as $\boldsymbol{\alpha}$ decreases, $\boldsymbol{\beta}$ increases.

Step 2: Significance Level α

- Type I Error (False Positive)

  α = P(reject H_0 | H_0 is true)

  - Reject H_0 when it is actually **true**.
  - The significance level $\boldsymbol{\alpha}$ represents the probability of making a Type I error (e.g., 0.01, 0.05, 0.10) → <u>value of α can be controlled</u>.
  - Applications of hypothesis testing that *only control* for the Type I error α are often called significance tests.
  - Example: Claiming that the new CAD system is effective (reduce response time $\mu < 25$) when in fact it does not. (Note: The claim is based on H_1.)
- Type II Error (False Negative)

  β = P(fail to reject H_0 | H_0 is false)

  - Fail to reject H_0 when it is actually **false**.
  - The probability of making a Type II error is $\boldsymbol{\beta}$, <u>cannot be specified in advance</u>. Why?
  - To avoid the risk of making a Type II error by concluding **"fail to reject $\boldsymbol{H_0}$"** instead of **"accept $\boldsymbol{H_0}$"**.
  - Example: Claiming that the CAD system is not effective (does not reduce the response time $\mu \geq 25$) when in fact it does. (Note: The claim is based on H_1.)

Step 2: Significance Level α

How β depends on the true population mean (<u>unknown</u>)?

In the CAD new system example:

$$H_0\colon \mu \geq 25,\ H_1\colon \mu < 25,$$

Think:

- If the true mean was **15**, then the sample mean will most likely be less than **25**. [less likely to NOT reject H_0]
- If the true mean is **24**, then the sample mean may or may not be less than **25** [more likely to NOT reject H_0]
- The **further away** the **true mean** from the **hypothesized value**, the **smaller** the value of $\boldsymbol{\beta}$.

Step 2: Significance Level α

Improving the Power of the Test $(1-\beta)$

- Power of test = $1-\beta$
  - is the probability of not committing a type II error.
  - should be high to allow us to make a valid conclusion.
- How to ensure sufficient power?
  - Power of test is sensitive to **sample size**
    - small sample sizes generally result in a low value of $1-\beta$
  - Power can be increased by taking larger samples
    - allows detection of small differences between sample statistics and population parameters with more accuracy
    - if α is small (a small level of significance), compensate by having a large sample size.

Step 3: Compute Test Statistic

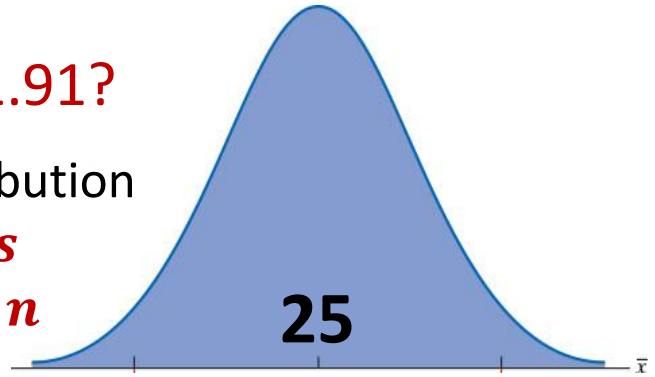
- The decision to reject or fail to reject a null hypothesis is based on computing a test statistic from the sample data.
- The test statistic used depends on the type of hypothesis test and certain assumptions about the population. <u>(fundamentally, sampling distribution of the test statistic)</u>
- Test statistic for one-sample hypothesis tests:

Type of Test	**Test Statistic**	
One-sample test for mean μ, $\boldsymbol{\sigma}$ **known**	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	$\sim N(0,1)$
One-sample test for mean μ, $\boldsymbol{\sigma}$ **unknown**	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$\sim t(n-1)$
One-sample test for proportion p	$z = \dfrac{\hat{p} - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}$	$\sim N(0,1)$

Step 3: Compute Test Statistic

Where is 21.91?

- z- or t-distribution
- Sample std $\boldsymbol{s}$
- Sample size $\boldsymbol{n}$



- In the CAD new system example, a sample of 44 customers revealed a mean response time of 21.91 minutes and a sample standard deviation of 19.49 minutes.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{21.91 - 25}{19.49/\sqrt{44}} = -1.05$$

- The test statistic $t = -1.05$ indicates that the sample mean of 21.91 is 1.05 standard errors below the hypothesized population mean of 25 minutes.
- Should we conclude that 21.91 is significantly lower than 25 – meaning the new system is effective?

Step 4&5: Decision Rule and Conclusion

- The decision to reject or fail to reject H_0 is based on comparing the value of the test statistic to a critical value from the sampling distribution of the test statistic when the null hypothesis is true, at the chosen significance level (α).
- What is critical value?
  - A threshold that separates the **rejection region**, where we reject H_0 and the **non-rejection region**, where we fail to reject H_0.
  - It is based on the significance level (α) and the sampling distribution of the test statistic (e.g. z or t-distribution).
  - It tells us how extreme a sample result must be before we decide that it's unlikely to occur by chance if H_0 were true.
- If the test statistic falls in the rejection region, we reject H_0; otherwise, we fail to reject it.

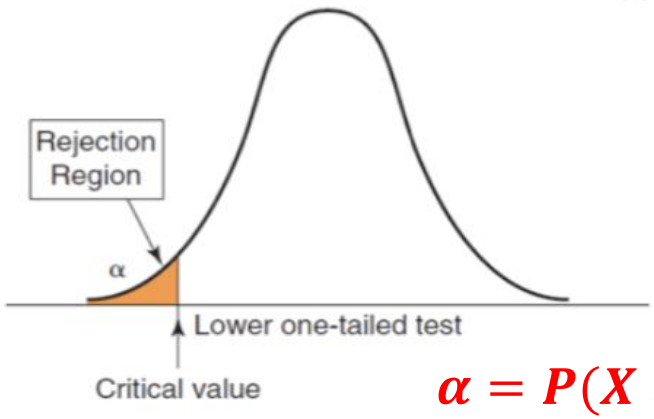
Step 4&5: Decision Rule and Conclusion

Question: If $\alpha = 0.05$,

- **What is z_α?**
- **What is $z_{1-\alpha}$?**

H_0: $\mu \geq \mu_0$

H_1: $\mu < \mu_0$

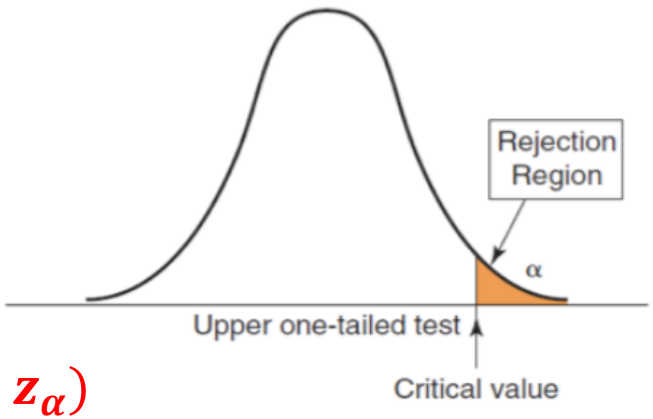


$\alpha = P(X \leq z_\alpha)$

Lower-tailed test

H_0: $\mu \leq \mu_0$

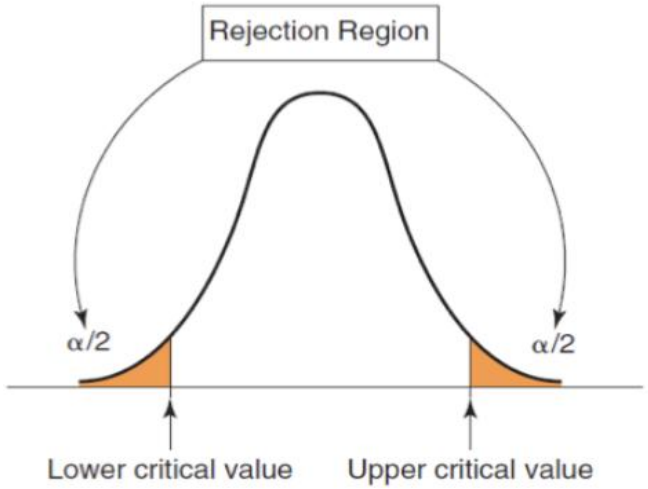
H_1: $\mu > \mu_0$



Upper-tailed test

H_0: $\mu = \mu_0$

H_1: $\mu \neq \mu_0$



Two-tailed test

$\alpha/2 = P(X \leq z_{\alpha/2})$

For a one-tailed test,

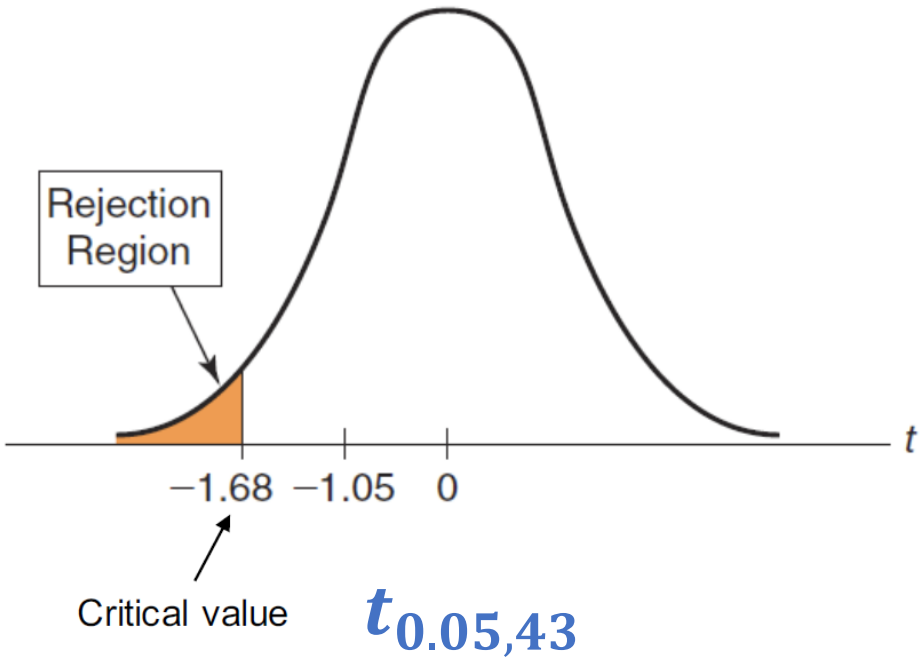
- If H_1 is stated as $<$, the rejection region is in the lower tail;
- If H_1 is stated as $>$, the rejection region is in the upper tail;

(Just think of the <u>inequality as an arrow pointing to the proper tail direction</u>.)

Step 4&5: Decision Rule and Conclusion

CAD New System Example (continued)

- Finding the critical value and drawing a conclusion
- $t = -1.05$ does not fall in the rejection region.
- Fail to reject H_0.



Even though the sample mean of 21.91 is less than 25, we are **unable to conclude** that the true population mean is less than 25 minutes as the sampling error is too large (e.g., sample Std is 19.49).

Step 4&5: Decision Rule and Conclusion

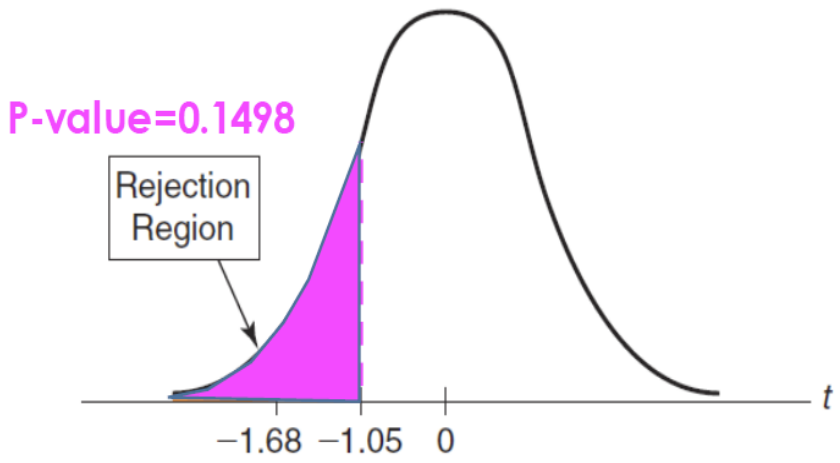
- An alternative approach of any hypothesis test (setting up a decision rule) uses the p-value rather than the critical value.
- A ***p*-value** is the probability, assuming H_0 is true, of obtaining a sample result **as extreme as or more extreme than** the one actually observed.
- A p-value is the observed significance level, measures how strong the evidence is against H_0.

  p-value = $P(X \leq \text{test statistic})$ (X's distribution: z, t, etc.)
- The p-value decision rule is to:

  Reject H_0 if p-value $\leq \alpha$
- Example:

  p-value = $\mathrm{P}(t \leq -1.05)$ = 0.1498, df=43

→ Fail to reject H_0 because p-value (0.1498) is not less than α (0.05)



Two-Tailed Hypothesis Test for Population Mean

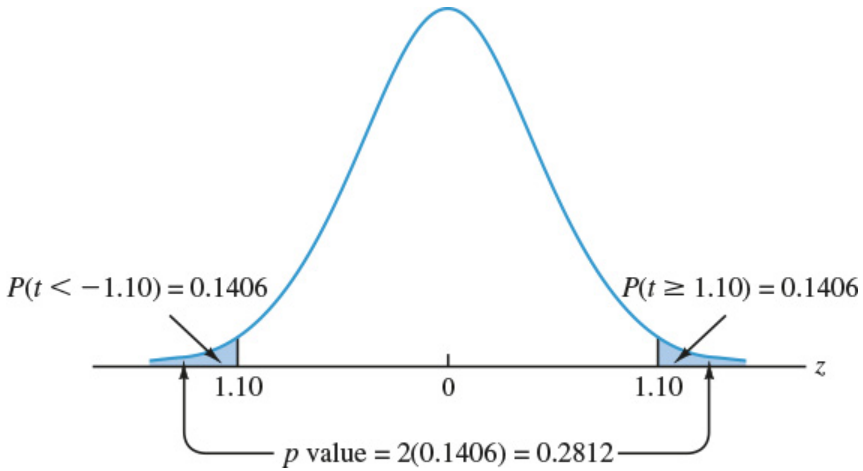
- Holiday Toys' marketing director surveys a sample of $n = 25$ retailers to test whether a new product per store demand is $\mu_0 = 40$ units.
- The director sets the hypotheses as a two-tailed test, with $\alpha = 0.05$.

$$H_0\colon \mu = 40$$
$$H_a\colon \mu \neq 40$$

- With $\bar{x} = 37.4$ and $s = 11.79$, we have

$$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} = \frac{37.4-40}{11.79/\sqrt{25}} = -1.10$$

$$p\text{-}value = \mathbf{2}P(t \leq -1.10) = 0.2812$$



- Because $p\text{-}value > \alpha$, H_0 cannot be rejected, and the director does not have enough evidence to conclude that per store demand is 40 units.

Double the probability (or tail area) to obtain the p-value

Taken Together – Logic of Hypothesis Testing

- Proof by Contradiction (反证法): (1) Assume H_0 is true, (2) Find evidence to reject H_0.

Example: A lower tailed hypothesis test, $H_0: \mu \geq 25$, $H_1: \mu < 25$ (the CAD new system example)

1. Assume H_0 is true $\rightarrow$ the population mean $\mu \geq 25$, could be 25, 26, 50, 1000 ...
2. Let us focus on the case where population mean μ=25, why this value? (Think: if we reject (or fail to reject) H_0 when $\mu = 25$, would we also reject (or fail to reject) H_0 when $\mu > 25$, such as $\mu = 26$?)
3. Ask: If $\mu = 25$, how likely is it to observe our sample mean $\bar{x}_1 = 21.91$?
   - How do we mean by "likely to observe"?
   - The sample mean $\bar{x}$ is a random variable, and its sampling distribution shows which values are likely or unlikely to occur.
   - For a normal distribution, about 95% of values fall within ± 1.960 standard errors of the mean – values outside this range are rare (小概率事件)!
   - If our observed sample mean (21.91) is too far from 25 –beyond the typical 95% range – it is unlikely to come from a population with mean 25.
   - We can then say we are 95% confident the population mean is not 25.
   - Or equivalently, we reject H_0 at the significance level 5%.

Taken Together – Logic of Hypothesis Testing

4. However, since we don't know the population standard deviation (σ), how can we determine whether the sample mean of 21.91 falls below the 95% critical boundary, such as $25 - 1.960\sigma$?

- Note: the random variable $\bar{x} \sim N\left(\mu, \left(\sigma/\sqrt{n}\right)^2\right)$, or $\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$, but, we do not have $\bar{x} \sim N\left(\mu, \left(s/\sqrt{n}\right)^2\right)$

5. Luckily, we have $\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim t_{n-1}$, and when $n > 30$, we have $\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim N(0,1)$, t-distribution is very similar to z-distribution, $t_{\alpha/2} \approx z_{\alpha/2}$.

6. Compare the test statistic to a critical value.

- If the observed $\bar{x}_1$ is too extreme (falls in the rejection region, e.g., below the 5% lower tail),→ this sample is unlikely to come from a population where μ=25. → We reject H_0.
- Notice that: $\frac{\bar{x}-\mu}{s/\sqrt{n}} < -z_{\alpha/2} \rightarrow \bar{x} < \mu - z_{\alpha/2}\frac{s}{\sqrt{n}}$ familiar? confident interval!
- Note the differences: $z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ $z_{\alpha/2}\frac{s}{\sqrt{n}}$ $t_{\alpha/2}\frac{s}{\sqrt{n}}$

Taken Together – Logic of Hypothesis Testing

- **Intuition:**
  - If something would be *very unlikely* under the assumption that H_0 is true,
    → we have **evidence against** H_0.
  - This is why hypothesis testing follows the logic of **反证法 (proof by contradiction)** — we assume H_0 is true and see whether the data contradict it.
- Based on what we have learned, let use revisit a few key points:
  - Why do we set the null hypothesis (H_0) as the statement we aim to reject?
  - Why do we say "fail to reject" rather than "accept"?
  - Why does the equality sign always appear in H_0?
  - Why the equality part of the hypotheses is always in the null hypothesis.
  - Why is a confidence interval written as "point estimate ± critical value × standard error"?
  - Why compute the test statistic $\frac{\bar{x}-\mu}{s/\sqrt{n}}$? Why is it sometimes called a z-score (z-test) and sometimes a t-score (t-test)?
  - How are confidence intervals and hypothesis testing connected?
  - Beyond the normal distribution case, → The same reasoning applies to any distribution, as long as we know its **critical value**, $P(Y \leq y_\alpha) = \alpha$ for a significance level α (for the lower tailed test).

Summary of Hypothesis Tests About a Population Mean

Step 1. Develop the null and alternative hypotheses

Step 2. Specify the level of significance

Step 3. Collect the sample data and compute the value of the test statistic

Step 4. Use the value of the test statistic to compute the *p*-value, or test statistic.

Step 5. Reject H_0 if *p*-value $\leq \alpha$, or test statistic $\leq$ ($\geq$) critical value for lower (upper) tail test

Step 6. Interpret the statistical conclusion in the context of the application

	Lower Tail Test	**Upper Tail Test**	**Two-Tailed Test**
Hypotheses	$H_0\colon \mu \geq \mu_0$ $H_a\colon \mu < \mu_0$	$H_0\colon \mu \leq \mu_0$ $H_a\colon \mu > \mu_0$	$H_0\colon \mu = \mu_0$ $H_a\colon \mu \neq \mu_0$
Test Statistic	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$
***p*-Value**	$= T.DIST(t, n-1, TRUE)$	$= 1 - T.DIST(t, n-1, TRUE)$	Double the minimum value of the two one-tail cases